

THE HILBERT SERIES OF THE INVARIANTS OF TERNARY SEPTICS, OCTICS AND NONICS

ANONYMOUS

ABSTRACT. We determine the Hilbert series of the rings of $\mathrm{SL}_3(\mathbb{C})$ -invariants of ternary forms of degrees 7, 8 and 9. Within the literature we found, the largest degree previously completed was 6 (Bedratyuk–Xin, 2011).

- The septic series equals the candidate R_7 of version 0.1.0 of this work, where it was proved only through degree 760 and conditionally beyond. Its least denominator has degree 1386.
- The octic series has a least denominator of degree 2331 and a palindromic numerator of degree 2286.
- The nonic series has a least denominator of degree 1393 and a palindromic numerator of degree 1338.

The method is known. Derksen’s universal denominators bound the poles of the series by torus data, the functional equation fixes the degree of the numerator, and finitely many exact coefficients then determine the series, as in Makam’s work on matrix semi-invariants. Our contributions are:

- the exhaustive torus-bound computations for $d = 7, 8, 9$, reproduced by a separately written program;
- correctness statements for the two coefficient engines;
- the exact coefficients, with archived per-prime residues;
- certificates, validated by re-deriving the known series for $d = 5, 6$;
- structural consequences: exact numbers of minimal generators in low degrees, and constraints on the degrees of any homogeneous system of parameters.

This version also corrects two attribution omissions in version 0.1.0.

1. INTRODUCTION

Let $I_{3,d} = \mathbb{C}[S^d\mathbb{C}^3]^{\mathrm{SL}_3}$ and let $H_d(t) = \sum_n a_n t^n$ be its Hilbert series, graded by degree.

- The series is classical for $d \leq 4$; see [2] and, for the history and methods up to 1991, Broer [3].
- Bedratyuk and Xin [2] computed $d = 5, 6$ by MacMahon partition analysis.
- For $d = 7$, Bedratyuk [1] gave the coefficients through degree 21.

We found no complete series for any $d \geq 7$. The search record is in `PRIOR_ART.md`; MathSciNet and zbMATH were not searched.

Version 0.1.0 [12] gave a rational function R_7 and proved $H_7 = R_7$ through degree 760. Full equality was proved only under an unverified hypothesis on a homogeneous system of parameters (hsop).

Theorem 1 (computer-assisted). (1) $H_7 = R_7$. *The least denominator of H_7 is a product of cyclotomic polynomials of degree 1386 (Table 2), and the numerator is palindromic of degree 1350.*

(2) $H_8 = N_8/D_8$, where $\deg D_8 = 2331$, and N_8 is palindromic of degree 2286.

(3) $H_9 = N_9/D_9$, where $\deg D_9 = 1393$, and N_9 is palindromic of degree 1338.

The polynomials are in `data/*/d{7,8,9}_rational.json`. The first terms are:

$$H_8 = 1 + t^3 + 6t^6 + 81t^9 + 1990t^{12} + 46543t^{15} + \dots, \quad H_9 = 1 + t^4 + 6t^6 + 3t^7 + 46t^8 + 118t^9 + 612t^{10} + \dots$$

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Method and what is new. Each ingredient of the method is known.

- Derksen [4] proves that the universal denominator of a graded module M is $\prod_r \Phi_r^{m_r}$ with $m_r = \dim M/I^{[r]}M$ [4, Thm. 1.10], where $I^{[r]}$ is generated by the homogeneous elements whose degree is not divisible by r . The universal denominator is the least common multiple of the denominators of all submodules.
- Derksen also proves that for connected reductive G with maximal torus T , $\text{udenom}(S^G)$ divides $\text{udenom}(S^T)$ [4, Thm. 3.4], and gives $\text{udenom}(S^T)$ combinatorially [4, Thm. 5.1].
- Knop’s criterion fixes the degree of H_d , and the rings are Gorenstein. Hence $H_d \cdot B$ satisfies a signed reciprocity for the denominators B used below.
- Consequently half the coefficients determine the series. Makam used this combination for matrix semi-invariants [8, §§3,6].

Our contributions are the new cases and what makes them auditable:

- exhaustive torus-bound computations (§4), reproduced by a separately written program;
- correctness statements for the coefficient engines (§3);
- the exact data, with archived per-prime residues;
- the certificates;
- the consequences in §6.

Corrections to version 0.1.0. Version 0.1.0 is immutable; we record two attribution omissions here. Neither affects any stated result.

- (1) Its §7 describes an a priori denominator for H_7 as available “by standard means”, without citing Derksen [4], who gives the sharper theory used here.
- (2) It presents the combination of a denominator theorem, the functional equation and finitely many exact coefficients as part of its methodological contribution. This is Makam’s strategy [8, §6], which [12] cites only for Knop’s criterion.

2. THE FINITE-DETERMINATION CERTIFICATE

Let $G = \text{SL}_3$, $V = S^d \mathbb{C}^3$, $A = \mathbb{C}[V]^G$ and $\delta = \dim V - 8$, the Krull dimension. We use:

- $\dim V = 36, 45, 55$ and $\delta = 28, 37, 47$ for $d = 7, 8, 9$.
- A is Cohen–Macaulay [6]. It is factorial because SL_3 has no nontrivial characters, and hence Gorenstein [9].
- $S^d \mathbb{C}^3$ is not coregular for $d \geq 4$ [10]. Being an irreducible module of a connected simple group, it is therefore 2-large [5, Thm. 3.7].
- Knop’s criterion [7], as stated in [8, Thm. 3.3], gives $\deg H = -\dim V$.
- Stanley’s theorem [11] then gives $H(1/t) = (-1)^{\delta} t^{\dim V} H(t)$.

Proposition 2. *Let $T \subset G$ be the diagonal torus, write $\text{udenom}(\mathbb{C}[V]^T, t) = \prod_r \Phi_r^{b_r}$, and let $B = \prod_r \Phi_r^{p_r}$ with $p_r = \min(\delta, b_r)$. Then $P = H \cdot B$ is a polynomial satisfying $P(t) = \varepsilon t^K P(1/t)$, where $K = \deg B - \dim V$ and $\varepsilon = (-1)^{\delta+p_1}$. Consequently H is determined by $a_0, \dots, a_{\lfloor K/2 \rfloor}$.*

Proof. The pole order of H at a primitive r -th root of unity is at most the exponent of Φ_r in $\text{udenom}(A)$, because A is a submodule of itself. By [4, Thm. 3.4], that exponent is at most b_r . It is also at most δ : by homogeneous Noether normalisation A is finite over a polynomial ring in δ variables, so Hilbert–Serre applies.

Hence $H \cdot B$ has no finite poles. A rational function with no finite poles is a polynomial. The signed reciprocity follows from the functional equation together with $B(1/t) = (-1)^{p_1} t^{-\deg B} B(t)$. A polynomial of degree at most K with this symmetry is determined by its coefficients of degree at most $\lfloor K/2 \rfloor$. $\square$

For $d = 7, 8, 9$ the sign is $\varepsilon = +1$. Table 1 gives the sizes involved.

TABLE 1. Certificate sizes. “Needed” is $\lfloor K/2 \rfloor$; “exact data” is the range of the archived exact coefficients; “modulus” is the product of the distinct primes, compared with $2^{\binom{n+\dim V-1}{\dim V-1}}$ at $n = \lfloor K/2 \rfloor$, which bounds $2a_n$.

d	$\deg B$	K	needed	exact data	modulus (bits)	$2 \times \text{bound}$ (bits)
7	1452	1416	708	0–760 (weight counting)	240	201
8	2433	2388	1194	0–1194 (grid)	372	272
9	1427	1372	686	0–690 (grid)	372	276

3. EXACT COEFFICIENTS

Two independent algorithms compute $a_n \bmod p$; the Chinese remainder theorem then gives the integers. Throughout, $W = \{(a - c, b - c) : a + b + c = d\} \subset \mathbb{Z}^2$ is the set of torus weights of V , one for each monomial $x^a y^b z^c$.

Proposition 3 (grid engine; `scripts/mw5.c`, `mw6.c`). *Let $L \geq 0$ and $M > dL + 4$. Let p be a prime with $M \mid p - 1$, and let $\omega \in \mathbb{F}_p$ have exact order M . For $(i, j) \in (\mathbb{Z}/M)^2$ set $e_3 = -i - j$, $E = (i, j, e_3)$, and*

$$w(i, j) = \prod_{a \neq b} (1 - \omega^{E_a - E_b}), \quad F_{i,j}(t) = \prod_{(u,v) \in W} (1 - \omega^{ui+vj} t)^{-1} \in \mathbb{F}_p[[t]].$$

Then for every $n \leq L$,

$$a_n \equiv M^{-2} \sum_{\substack{0 \leq i < j < e_3 < M \\ i+j+e_3 \equiv 0}} w(i, j) [t^n] F_{i,j}(t) \pmod{p}.$$

Proof. By the Weyl integration formula, $H(t) = \frac{1}{6} \text{CT}_z [\prod_{\alpha} (1 - z^{\alpha}) \prod_{w \in W} (1 - tz^w)^{-1}]$. Here $z = (z_1, z_2)$, $z_3 = (z_1 z_2)^{-1}$, the product over α runs over the six roots z_a/z_b , and CT takes the constant term of each t -coefficient.

- (1) *Exponent bound.* The coefficient of t^n is an integer Laurent polynomial whose exponents have absolute value at most $dn + 4$: the weights contribute at most dn and the Weyl numerator at most 4.
- (2) *No aliasing.* For a monomial z^m with $|m_1|, |m_2| < M$, the average $M^{-2} \sum_{(i,j)} \omega^{m_1 i + m_2 j}$ equals 1 if $m = 0$ and 0 otherwise, since ω has exact order M and M is invertible mod p . So the average over the order- M grid computes the constant term exactly modulo p whenever $M > dL + 4$.
- (3) *Orbit reduction.* The integrand is invariant under S_3 permuting (E_1, E_2, E_3) , and it vanishes when two E_a coincide. So the grid sum is 6 times the sum over representatives $i < j < e_3$, which cancels the factor $\frac{1}{6}$.
- (4) *Power series.* The engine builds $F_{i,j}$ one factor at a time using $s_n \leftarrow s_n + x s_{n-1}$, which is multiplication by $(1 - xt)^{-1}$.

□

Code map for mw5.c.

- Line 24 checks $M \mid p - 1$ and $p < 2^{62}$. Line 25 enforces $M \geq 2(dL + 2) + 1 > dL + 4$.
- Lines 29–31 choose ω and verify its exact order M by testing $\omega^{M/q} \neq 1$ for every prime $q \mid M$.
- Line 51 forms $w(i, j)$. Line 49 selects $i < j < e_3$. Line 56 forms ω^{ui+vj} . Lines 57–58 perform the recurrence.
- Line 65 multiplies by M^{-2} .

- Arithmetic is Montgomery reduction with 128-bit products for $p < 2^{62}$. The engine refuses weight sets with more than 64 elements (line 35), so $d \leq 9$.
- `mw6.c` is the same algorithm for $p < 2^{31}$.

Proposition 4 (weight counting; `methodAd.c`, `methodA16.c`). *Let $f_n(A, B)$ be the number of multisets of n elements of the multiset of GL_3 -weights $\{(a, b, c) : a+b+c = d\}$ with $\text{sum}(A, B, dn - A - B)$. If $3 \nmid dn$ then $a_n = 0$. Otherwise, with $k = dn/3$,*

$$a_n = f_n(k, k) - f_n(k-1, k+1) - f_n(k, k-1) - f_n(k-2, k) + f_n(k-1, k-1) + f_n(k-2, k+1).$$

Proof. a_n is the multiplicity of $\det^k$ in $S^n(S^d V)$ as a GL_3 -module. By the Weyl character formula this multiplicity is $\sum_{w \in S_3} \text{sgn}(w) \text{mult}(w(\lambda + \rho) - \rho)$, with $\lambda = (k, k, k)$ and $\rho = (2, 1, 0)$. The six terms are those displayed.

The program computes f_n by one unbounded-knapsack pass per weight, keeping only the coordinates at most $dn/3 + 2$. This is enough because weights are nonnegative and every target coordinate is at most $k + 2$. Arithmetic is modulo a prime $p < 2^{16}$, which the program checks, with 16-bit storage. $\square$

Data.

- *Septics*: exact a_n for $n \leq 760$ by Proposition 4 at 15 distinct 16-bit primes. The archived per-prime outputs are re-lifted on every replay, and `FULL=1` regenerates them in about three minutes.
- *Octics and nonics*: exact a_n by Proposition 3 at six distinct 62-bit primes. For each case the six raw residue arrays are archived, and `certify_ternary.py` re-lifts them itself. It checks every congruence, the array lengths, the metadata and the bound, and compares with the frozen output before writing anything.
- The two algorithms share no code and no mathematics beyond the definition of a_n . They agree modulo two 16-bit primes for $n \leq 450$ ($d = 8$) and $n \leq 400$ ($d = 9$).

4. THE TORUS BOUNDS

By [4, Thm. 3.4 and its proof] and [4, Thm. 5.1], b_r is the largest dimension of $V_S//T$, where:

- $S = S(t, \zeta) = \{w \in W : \chi_w(t) = \zeta\}$ is the set of weights on which some $t \in T$ acts by some primitive r -th root of unity ζ ;
- V_S is the span of the corresponding monomials.

For finite $S \subset \mathbb{Z}^2$ put $\kappa(S) = \dim V_S//T$.

Lemma 5. $\kappa(S) = |S_0| - \text{rank } S_0$, where $S_0 = \{w \in S : -w \in \text{cone}(S)\}$. In particular:

- κ is monotone under inclusion;
- $\kappa(S) = 0$ if S lies in an open half-plane through 0 (for example, on an affine line not through 0);
- $\kappa(\{0\}) = 1$, and $\kappa(\{w\}) = 0$ for $w \neq 0$.

Proof. $\mathbb{C}[V_S]^T$ is the monoid algebra of the nonnegative integer relations $\sum c_w w = 0$, so its dimension is that of the relation cone. That cone lies in $\mathbb{R}^{S_0}$ and contains a relation that is strictly positive on S_0 . So it has the dimension of $\ker(\mathbb{R}^{S_0} \rightarrow \mathbb{R}^2)$, which is $|S_0| - \text{rank } S_0$. The remaining statements follow directly. $\square$

Lemma 6 (exhaustive enumeration). *Every level set $S = W \cap \phi^{-1}(\zeta)$ of a homomorphism $\phi : \mathbb{Z}^2 \rightarrow \mathbb{C}^*$ at a root of unity ζ of exact order r is contained in one of the following sets, with the same r :*

- (i) Rank two. *If the differences of S span a rank-2 lattice, then $S = W \cap (w_0 + \Lambda)$, where $\Lambda = \ker \phi$ is the kernel of $(x, y) \mapsto cx + ey \bmod N$ for some N at most the largest $|\det|$ of two weight differences (147, 192, 243 for $d = 7, 8, 9$). One may take $c \mid N$. Here r is the order of w_0 in $\mathbb{Z}^2/\Lambda \cong \mathbb{Z}/N$.*
- (ii) Rank one. *If S lies on a line through 0 with primitive direction u , then $S \subseteq \{ku : k \equiv \rho \pmod{m}\}$ for some m at most the diameter of the line's weight set, with $r = m/\gcd(m, \rho)$. If S lies on a line not through 0, then $\kappa(S) = 0$.*
- (iii) One element. *If $S = \{w\}$, then $\kappa = 0$ for $w \neq 0$. For the zero weight, which occurs when $3 \mid d$, $S = \{0\}$ has $\kappa = 1$ and $\zeta = \phi(0) = 1$, so $r = 1$.*

Consequently $b_r = \max \kappa$ over the finitely many sets listed. For every r not attained with $\kappa > 0$, $b_r = 0$.

Proof. (i) $\ker \phi$ contains the rank-2 lattice of differences, so its index divides that lattice's index. It is therefore at most the $|\det|$ of any two independent differences. $\mathbb{Z}^2/\ker \phi$ embeds in $\mathbb{C}^*$, so it is finite cyclic and ϕ induces an isomorphism onto μ_N . A primitive map (a, b) can be multiplied by a unit k with $ka \equiv \gcd(a, N)$, which gives the normal form $c \mid N$.

(ii) Points ku lie in S exactly when $\phi(u)^k = \zeta$. If $\phi(u)$ has infinite order, then $|S| \leq 1$. If it has order m , the condition is a congruence class, and classes of size at least 2 require m to be at most the diameter. On an affine line not through 0 a linear form is positive, so $\kappa = 0$ by Lemma 5.

(iii) This is Lemma 5 together with $\phi(0) = 1$.

Monotonicity of κ gives the conclusion. $\square$

`scripts/pole_bounds.py` implements Lemma 6. A second program, written blind by a separate agent (`independent/blind_torus_bounds/`), evaluates the lattice condition of [4, Thm. 5.1] directly. It maximises $\kappa(S)$ over subsets $S = W \cap (w_0 + L)$ with $r \mid o(S)$, where $o(S)$ is the order of $(0, 1)$ modulo the lattice generated by the $(w, 1)$, $w \in S$. The two programs give identical b_r for $d = 5, \dots, 9$. For the known cases $d = 5, 6$, b_r is at least the true multiplicity everywhere and positive exactly on the true pole set.

5. THE THREE SERIES

Septics. For $d = 7$, Proposition 2 requires $a_0, \dots, a_{708}$, which are exact. The script `certify_septic_uncondition` regenerates the bounds and checks three things in exact arithmetic:

- the least denominator D_7 of R_7 divides B ;
- $Q = R_7 B$ is a polynomial of degree at most 1416 with $Q(t) = t^{1416} Q(1/t)$;
- the series of R_7 equals the exact a_n for $n \leq 708$.

So $H_7 B = Q$, and $H_7 = R_7$.

Octics and nonics. No candidate is needed. The script `certify_ternary.py` proceeds as follows:

- (1) it re-lifts the archived residues;
- (2) it forms the lower half of $P = H \cdot B$ and completes it by reciprocity;
- (3) it reduces P/B to lowest terms;
- (4) it checks that the pole order at $t = 1$ equals δ .

The octic determination uses a_n for $n \leq 1194$. The nonic determination uses $n \leq 686$ of the archived range 0–690.

Three kinds of check lie outside the determinations:

- predictions modulo fresh 31-bit primes up to degree 1600 ($d = 8$) and 1000 ($d = 9$);
- the weight-counting agreement stated in §3;

- re-derivation by the same pipeline of Bedratyuk and Xin's H_5 and H_6 from fresh grid data, which reproduces their rational functions exactly.

TABLE 2. Multiplicities m_r of Φ_r in the least denominators ($r:m_r$; others 0). An asterisk marks orders where the torus bound $\min(\delta, b_r)$ exceeds m_r ; it always exceeds it by exactly 2.

$d = 7$	1:28, 2:16*, 3:28, 4:8*, 5:7*, 6:16*, 7:3, 8:4, 9:11*, 10:4, 11:2, 12:8*, 13:2, 14:1, 15:7*, 16:1, 17:1, 18:6*, 19:1, 20:1, 21:3, 22:1, 23:1, 24:4, 25:1, 26:1, 27:3, 29:1, 30:4, 33:2, 36:3, 39:2, 42:1, 45:2, 48:1, 51:1, 54:2, 57, 60, $\dots$, 78 (step 3):1, 87:1, 90:1, 108:1
$d = 8$	1:37, 2:16*, 3:37, 4:8*, 5:9*, 6:16*, 7:7*, 8:4, 9:14*, 10:4, 11:2, 12:8*, 13:2, 14:3, 15:9*, 16:1, 17:2, 18:6*, 19:1, 20:1, 21:7*, 22:1, 23:1, 24:4, 25:1, 26:1, 27:4, 28:1, 29:1, 30:4, 31:1, 33:2, 35:1, 36:3, 37:1, 39:2, 41:1, 42:3, 45:3, 48:1, 49:1, 51:2, 54:2, 57:1, 60:1, 63:2, 66:1, 69:1, 72:1, 75:1, 78:1, 81:1, 84:1, 87:1, 90:1, 93:1, 105:1, 108:1, 111:1, 123:1, 126:1, 147:1
$d = 9$	1:47, 2:26*, 3:16, 4:14*, 5:11*, 6:8, 7:9*, 8:8*, 9:4, 10:6, 11:3, 12:4, 13:3, 14:5, 15:3, 16:4, 17:2, 18:2, 19:2, 20:3, 21:2, 22:2, 23:2, 24:2, 25:2, 26:1, 27:1, 28:2, 29:1, 30:1, 31:1, 32:2, 33:1, 34:1, 35:1, 36:1, 37:1, 38:1, 40:1, 41:1, 42:1, 43:1, 44:1, 46:1, 47:1, 48:1, 49:1, 50:1, 55:1, 56:1, 64:1

6. WHAT THE SERIES DETERMINE

The series constrain the structure of the invariant rings. The following statements are consequences of Theorem 1 alone. The script `scripts/consequences.py` computes them, and the output is in `data/consequences.json`.

Minimal generators in low degree. Write $g_n = \dim(A_+/A_+^2)_n$ for the number of minimal generators of degree n . Then $g_n = a_n - \dim(A_+^2)_n$. The dimension $\dim(A_+^2)_n$ is at most the number M_n of degree- n monomials in the lower-degree generators. When $M_n \leq 1$ it equals M_n , because A is a domain and a product of nonzero elements is nonzero. This gives the following:

- $d = 7$: $g_6 = 3$ and $g_9 = 13$, while $415 \leq g_{12} \leq 420$.
- $d = 8$: $g_3 = 1$ and $g_6 = 5$ (the only decomposable in degree 6 is the square of the cubic), while $75 \leq g_9 \leq 80$.
- $d = 9$: $g_4 = 1$, $g_6 = 6$, $g_7 = 3$, $g_8 = 45$ and $g_9 = 118$, while $606 \leq g_{10} \leq 611$.

Systems of parameters. If A is finite over $\mathbb{C}[f_1, \dots, f_\delta]$ with $\deg f_i = k_i$, then the least denominator divides $\prod_i (1 - t^{k_i})$. So for every r at least m_r of the k_i are divisible by r . Consequences:

- every hsof of $I_{3,7}$ has a member whose degree is divisible by 108, one divisible by 90, one divisible by 87, and so on;
- for $I_{3,8}$ there is a degree divisible by 147;
- for $I_{3,9}$ there is a degree divisible by 64.

Minimising $\sum_i k_i$ subject to these constraints, with each k_i a degree in which invariants exist, is an integer program. It gives $\sum_i k_i \geq 1428, 2391$ and 1457 for $d = 7, 8, 9$. For septics the minimum is attained by the multiset K of [12, Thm. 2]. Whether an hsof with minimal degree sum exists is not determined by the series.

Secondary invariants. Let $c_d = \lim_{t \rightarrow 1} (1 - t)^\delta H_d(t)$. Since A is Cohen–Macaulay, it is free over any hsof, and the number of secondary invariants is $c_d \prod_i k_i$. The exact values of c_d are in `data/consequences.json`; for example $c_7 \approx 3.3029 \cdot 10^{-10}$.

What is not determined. The series do not determine the generators themselves, the degrees of an actual hsof, the relations beyond the counts implied above, or whether the rings are complete intersections. They do provide exact targets for all such computations.

TABLE 3. Checks. “Exact” means integer equality; “mod p ” means agreement modulo the stated prime(s), which is evidence and not an integer certificate.

Check	Result
Torus bounds b_r , $d = 5, \dots, 9$: two separately written programs	identical
Torus bounds versus the known least denominators, $d = 5, 6$	$b_r \geq m_r$; same support
Method re-derives Bedratyuk–Xin’s H_5 , H_6 from fresh grid data	exact, equal
H_7 : certificate with exact a_n , $n \leq 708$ (weight counting, 15 primes)	exact
H_8 : re-lift of six archived residue arrays; determination; pole order 37	exact
H_9 : re-lift of six archived residue arrays; determination; pole order 47	exact
H_8 , H_9 : weight counting modulo 65521, 65519 ($n \leq 450, 400$)	mod p , agree
H_8 , H_9 : prediction modulo a fresh 31-bit prime ($n \leq 1600, 1000$)	mod p , agree

7. VERIFICATION

Review record (provenance, not a mathematical test):

- the septic argument was reviewed adversarially by a model from another vendor (GPT-5.6) in two rounds: round 1 found write-up gaps, and round 2 gave “valid with minor gaps”, which were repaired;
- version 0.2.0 was reviewed by an external model review, whose required revisions are made here (see [review/](#)).

8. LIMITATIONS

- All three theorems are computer-assisted. They rest on the cited theorems, on Propositions 3 and 4 and their implementations, and on the enumeration of Lemma 6.
- For octics and nonics, the determining coefficients come from one algorithm, the grid engine. The second algorithm, weight counting, checks them only modulo primes and to lower degrees.
- No step is formally verified. Knop’s criterion is used through Makam’s statement. All review is model-based.
- The next case, $d = 10$, has 66 weights. The present engines accept at most 64, so it needs an implementation change and renewed checking. With the torus bound it would need exact a_n up to degree 3249, roughly 100 CPU-hours with the present algorithm. The recurring excess of exactly 2 in the torus bounds (Table 2) suggests that sharper bounds, for example from larger centralisers, would reduce this cost.

PROVENANCE AND AI-USE STATEMENT

This work was produced by an AI research agent (Claude, Anthropic) under the direction of the publisher, Evidence Press. The problems, computations, proofs and text are AI-generated, and no human mathematical contribution is claimed. Parts of the verification were written by separate AI agents working from the definitions alone. No human specialist has checked the work.

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